INCH-POUND

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MILITARY SPECIFICATION

ELECTRO-EROSION MACHINES, ELECTRIC DISCHARGE "C" FRAME, RAM/QUILL TYPE, STATIONARY/ROTARY SPINDLE, MANUAL AND COMPUTER NUMERICAL CONTROL (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a "C" frame, ram/or quill, stationary/or rotating spindle type electric discharge machine with a manual or computerized numerical control (CNC) and external power supply unit with equipment and accessories specified herein. The requirements set forth in this specification and the completed ordering data (paragraph 6.2) establish the Government's minimum requirement for each procurement.

1.2 Classification. The electrical discharge machine shall be of the following types and of the sizes listed in tables I and II. The types and sizes to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Manually controlled machine.

Type II - Computerized numerical controlled (CNC) machine.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3410

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, (ATTN: Mr. Stephen Lowell), Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific

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acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks only as Unassembled Gears. (DOD adopted)

AGMA 2000-A88 - Gear Classification and Inspection Handbook. Tolerances and Measuring Methods for Unassembled Spur and Helical Gears. (DOD adopted)

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues. (DOD adopted)

ANSI B93.114M - Pneumatic Fluid Power - Systems - Standards for Industrial Equipment and Machine Tool Applications.

ANSI/ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Deptartment, 1430 Broadway, New York, NY 10018-3363.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA-227 - One-Inch Perforated Tape. (DOD adopted)

EIA-232 - Data Terminal Equipment and Data Circuit-Terminating Equipment, Interface Between, Employing serial Binary Data Interchange. (DOD adopted)

EIA-244 - Character Code for Numerical Machine Control Perforated Tape. (DOD adopted)

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EIA-267 - Axis and Motion Nomenclature for Numerically Controlled Machines. (DOD adopted)

EIA-274 - Interchangeable Variable Block Data Contouring, Format for Positioning, and Contouring/Positioning Numerically Controlled Machines. (DOD adopted)

EIA-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape. (DOD adopted)

(Application for copies should be addressed to the Electronic Industries Association, Engineering Deptartment, Standards Orders, 2001 Eye Street, N.W., Washington, DC 20006-1899.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The electro-erosion machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. The electrical discharge machine shall have a ram/quill with stationary and/ or rotating spindle, electric axis feed motors and external power supply unit, complete with equipment

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and accessories. When specified (see 6.2), the design and construction of the machine shall be in accordance with either the U.S. Customary System of Units (US) or the International System of Units (SI).

3.2.1. Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. Unless otherwise specified (see 6.2), all dials used to indicate machine axes position or tool movement shall be graduated in increments of 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as dial is rotated through the zero position.

3.2.1.2 Digital readout system. When specified (see 6.2), a digital readout (DRO) system shall be installed on the X, Y, and Z axes of travel. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0005 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0002 inch, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch-pound and metric units. An input power source fluctuation of $\pm$10 percent shall not adversely affect the system operation.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.1.2.2 Electronic display console. The electronic display console for the digital readout shall contain a universal digital display with individual illuminated digits of not less than 0.43 inches in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and

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shall be readable from a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed to be readily accessible and easily read from the operator's normal work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of the installed cables shall prevent any interference with normal machine operation.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt or remanufactured products are allowed under this specification.

3.2.3 Energy efficiency. The electro erosion machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation is influenced by workpiece size and configuration, the procuring activity shall provide part data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During operation, service, transportation, or storage of the machine, it shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter and a low lubrication level warning light that is in clear view

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of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. To provide for the replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. Replacement parts shall be domestically available.

3.3 Construction. The electrical discharge machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only when specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000-A88 and AGMA 2001-B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be a hardened steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The

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equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The electrical discharge machine shall consist of not less than the base and column, saddle assembly, table and work tank assembly, ram/quill head assembly, dielectric system, pneumatic system, and/or coolant pump system, electrical system, external power supply unit, and all other components/systems necessary for the machine to perform as specified herein.

3.4.1 Base and column. The base and column shall be castings or stress relieved weldments. They shall be designed and engineered to support all machine components and withstand all static loads and dynamic forces produced by the machine when operating at maximum capacity. The base or tank attached thereto shall be used as a reservoir for the dielectric coolant. Doors or removable access covers shall be provided in the base to permit servicing the dielectric reservoir and its components. The base shall incorporate, either as a integral part or as a separate unit, provisions for mounting the saddle and table. Means for leveling the base shall be provided.

3.4.2 Saddle and table. The saddle and table shall be designed to withstand all static and dynamic loads imposed upon them during normal operation of the machine. The top of the saddle shall be provided with ways or guides suitable for mounting a sliding table. The table shall be mounted to the saddle, and travel on the saddle in a direction square to the movement of the saddle. The saddle shall be mounted on guides or ways provided on the base. Both the saddle and table of the type I machines shall be provided with manual feed mechanisms. The manual feed mechanisms shall consist of a precision feed screw and feed screw nut, a hand wheel, and a graduated dial conforming to 3.2.1.1. The feed mechanisms shall be constructed to such precision as to allow positioning the saddle to the accuracy specified in table III. Saddle and table slides having box or dovetail ways shall have adjustable gibs the full bearing length of the saddle and table ways to compensate for wear. When recirculating linear ball bushings on bar ways are used each bar way shall be fully supported throughout its length, and shall be hardened to not less than Rockwell C-60. Ball bushings shall be of the adjustable diameter type. The table shall have an adequate number of T-slots and shall conform to the requirements specified in ANSI B5.1.

3.4.2.1 Saddle and table clamping devices. The saddle and table shall be provided with a suitable number of locking/clamping devices. These devices shall be capable of securely holding the saddle and table in position immediately upon completion of a designated move. The locking/clamping devices shall be the positive acting type to prevent movement of the components when clamping or unclamping. Both the saddle and table of type I machines shall be provided with clamping devices that can be operated manually or automatically from the operators station. Type II machines shall be activated automatically either from a part programmed tape or the operator's console.

3.4.3 Table work tank. The table work tank shall be either an integral part of the work table or a separate unit that can be attached to the work table. The table work tank shall be manufactured from a metal that is not adversely affected by the dielectric coolant. The work tank shall have a removable front panel and/or swing out door to facilitate setup and removal of the workpiece and changing of the electrode. The removable front panel and/or swing out door shall

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be designed to prevent leakage of the dielectric coolant when attached in place to the work tank and shall be provided with a sealing material that is impervious to the dielectric coolant. The front panel and/or swing out door shall be held in operating position by quick opening clamps. The work tank shall have provisions for controlling the coolant level as well as for filling and emptying the work tank. Provisions shall be made for automatically disconnecting electric power to the electrode in the event the coolant level falls below a predetermined level.

3.4.4 Ram/quill housing. The ram/quill housing shall be a casting or a fabricated steel weldment designed and proportioned to house and support the ram/quill and spindle assembly. The housing shall have scraped ways, hardened and ground ways or recirculating ball bushings to support the ram/quill unit, and shall also contain an electrical-mechanical feed system. If the ram/quill and spindle assembly requires a counterbalance system, the system shall be housed in the column assembly of the machine.

3.4.4.1 Ram/quill and spindle assembly. The ram/quill shall be constructed as one piece from heat-treated alloy steel or cast iron and shall house either an integral rotating spindle assembly or have provisions for mounting a rotating spindle attachment. The integral spindle if supplied shall be mounted in the ram/quill with its bearings arranged to compensate for thermal expansion and shall be of sufficient size to perform machining operations within the parameters of the machine. The spindle shall be supported and held in axial and radial alignment by precision ball or roller bearings. Spindle rotation shall be performed by variable speed electric motors. The assembly shall have provisions to allow dielectric coolant to be pumped through passages drilled in the electrode. It shall have means provided for mounting the electrode and be electrically insulated from the assembly to prevent conduction of electrical current through the ways or bearings. The ram/quill shall have feed and traverse capabilities within the housing assembly and shall meet the alignment accuracies as specified in table III. Distances between the table surface and the electrode holder shall be not less than the requirements specified in table I and II.

3.4.4.2 Ram/quill feed system. A ram/quill feed system shall be provided. The feed system for type I machines shall consist of a gear train driven by an electric motor or servo motor coupled to a precision feed screw. The feed system for type II machines shall be direct-current (DC) permanent magnet or an alternating-current (AC) servo motor coupled to a precision ground ball screw. The electric motors shall be in accordance with 3.4.9.1. The feed system shall have all pertinent controls necessary to assure proper positioning of the electrode in relation to the workpiece. The control system shall be electrically interfaced with the power supply, and shall automatically feed the electrode to the workpiece at the proper rate to maintain the preset conditions of spark gap, metal removal rate, and surface finish. Means shall be provided to permit manual operation of the feed system while the machine is not operating or in the machining mode.

3.4.4.3 Ram/quill position and stop indicator. Type I machines shall be provided with either a DRO system or micrometer quill stop and dial type indicator. Unless otherwise specified (see 6.2), both the stop and dial indicator shall be graduated in increments of 0.001 inch. Either system shall be usable within the full range of the ram/quill stroke. The DRO or dial indicator shall indicate downward movement and be capable of stopping the ram/quill at any preset position. Ram/quill positioning of type II machines shall be controlled by the

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CNC unit, either from part programmed tape or from the operator's console and displayed on the cathode ray tube (CRT) screen.

3.4.5 Dielectric coolant system. The dielectric coolant system shall consist of a reservoir in the base or attached thereto, a fast filling and emptying system, a pump system supplying coolant to the electrode and workpiece, a filtering system, and necessary piping, valves, and electrical controls.

3.4.5.1 Fast filling and emptying system. A fast filling and emptying system shall be provided for the table work tank. This system shall allow for rapid loading and unloading of the workpiece, and adjustment or inspection of the electrode. The system shall be provided with either a high capacity pump or a pressurized reservoir capable of operating from a shop compressed air system.

3.4.5.2 Dielectric coolant pump system. The pump shall be powered by an electric motor of a continuous horsepower rating capable of delivering not less than 10 gallons per minute. The pump shall supply coolant to the electrode, workpiece and work tank. It shall not be adversely affected by the coolant or the abrasive particles in the coolant. It shall pick up coolant either from the work tank or the reservoir, discharge the coolant through a filtering system to the electrode holder and then to the electrode. When it is not desirable or practicable to pump coolant through the electrode means shall be provided to allow coolant to be applied directly to the electrode gap or workpiece. Pressure gauges shall be provided for monitoring the coolant at the pump outlet. Pressure relief valves shall be provided, when required, to prevent build up of excessive pressure.

3.4.5.3 Filtering system. The filtering system shall have replaceable or washable element type filters capable of removing all solids larger than 5 microns. When specified (see 6.2), a filter designed for removing all solids down to 5 microns shall be furnished. Pressure gauges or a low-pressure alarm shall be provided to indicate the pressure drop across the filter(s) to allow the operator to determine when the element(s) need cleaning or replacing. Filters shall be designed and located to facilitate servicing.

3.4.6 Automatic electrode changer. When specified (see 6.2), the machine shall have an automatic electrode changer with a magazine capable of storing not less than the electrode capacity required for the style machine specified. The magazine shall be of the rotary type or any other closed loop recirculating configuration with the capability of presenting a preprogrammed electrode to the electrode changer for exchange with the electrode in the holder. Electrode change shall be accomplished by means of a power actuated mechanism capable of either interchanging an electrode in the holder with an electrode selected from the electrode magazine or removing the electrode from the holder and then inserting a electrode from the magazine. The magazine shall permit convenient loading and unloading of electrodes during operational setup. The electrode changer shall be arranged for random selection which shall include the capability of reusing a electrode that has been returned to the magazine after a programmed operation.

3.4.7 Power supply and control unit. Unless otherwise specified (see 6.2), an electrical pulse-type power supply and control unit of complete solid-state design shall be furnished with each electrical discharge machine. The circuitry shall be of the solid-state microprocessor based type, mounted on replaceable plug

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in circuit boards. The power supply unit shall have a continuous output rating of not less than 25, 30, 50, 60, 75, 100, 120, or 200 amperes as specified (see 6.2). The power supply of the type II machines shall be interfaced to the CNC control unit. The unit shall be adequately ventilated and cooled to prevent overheating and premature failure of electrical/electronic components. A filtering system shall be supplied to remove impurities from the incoming air supply; the filters shall be either washable or throwaway type filters. The control unit shall provide control of the metal removal rate, over cut, surface finish, automatic gap spacing of the electrode and low electrode wear.

3.4.8 Pneumatic system. The pneumatic system, when provided shall conform to the requirements of ANSI B93.114M.

3.4.9 Electrical system. The electrical system shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Unless otherwise specified (see 6.2), the machine shall be initially wired for operation on 460 volts.

3.4.9.1 Motors. Motors shall meet the requirements of 3.4.9. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), AC motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.10 Worklight. Unless otherwise specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.11 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated erosion hours. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4.12 Computer numerical control system (CNC). The type II electrical discharge machine shall have a fully automatic solid-state, soft-wired, integrated circuit, microprocessor based-type numerical control. The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, electrode change, axis movement, and other part program directed functions for electro erosion machines of the type specified herein. Controlled axes shall be identified in accordance with EIA Standard 267. Control features shall include linear and circular interpolations, dwell control, buffer and part program storage, buffer and part program edit capabilities and control diagnostics to board level. The MCU shall direct machine functions from part program data stored in memory, manually input data (MDI) or from the media specified herein. The control shall initiate a halt or error signal should a fault condition occur in the control. All necessary executive program routines shall be furnished in the form of 1-inch, eight-channel, binary-coded decimal punched tape in conformance with EIA Standard 227 mini-cassette or floppy disk for controls having volatile

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memories. When executive programs are supplied in the form of erasable programmable read only memory (EPROM), complete documentation sufficient to interpret and duplicate shall be provided or one additional EPROM chip shall be provided by the manufacturer. A line voltage ±10 percent from normal shall not adversely affect the numerical control system. The control shall be capable of functioning in ambient temperatures ranging from 50 to 120 degrees Fahrenheit (F) and relative humidities ranging from 5 to 95 percent noncondensing. The control shall automatically shut down when internal operating temperatures exceed safe limits. The control unit resolution shall be 0.0001 inch or 0.001 mm for each linear axis. The control shall also provide the following:

a. A 9-, 12-, or 14-inch monochrome or multi-color cathode ray tube (CRT) shall be provided for data display.

b. Sequence number search.

c. Programmable and optional stop.

d. Programmable tape rewind.

e. Absolute/incremental programming.

f. Automatic tape code recognition, in conformance with EIA Standards 244 and 358 (when specified, see 6.2).

g. Dry run, no motion program test.

h. Axis motion status display.

i. Conversational programming, when specified (see 6.2).

j. Scaling, when specified (see 6.2).

k. Inch/metric data input.

l. Optional block skip.

m Cycle start/feed hold buttons.

n. Axis lock.

o. Backlash compensation.

p. Programmable flush selection.

q. EIA Standard 232 serial program data input interface.

r. Photoelectric tape reader capable of reading a minimum of 300 characters per second with standard feed and take up spools 5-1/4 inches in diameter, when specified (see 6.2).

s. Plane selection.

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t. Electrode length compensation.

u. Electrode radius compensation.

v. Electrode center compensation.

w. Linear axis jog for each axis, continuous and incremental modes.

x. Block delete.

y. Lead screw pitch error compensation.

z. Part program storage of not less than 100 feet, if different (see 6.2).

aa. Parametric programming, when specified (see 6.2).

bb. Control memory shall be either solid-state, magnetic-bubble memory, winchester hard disk/flexible disk drive combination units or single/double flexible disk drive units as specified (see 6.2).

cc. Standby battery power for controls with volatile memories shall be a minimum of 72 hours, unless otherwise specified (see 6.2).

dd. Buffer storage: The buffer storage shall store transferred command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine functions.

ee. Reference zero (all axes home position).

ff. Floating zero.

gg. Adaptive control, when specified (see 6.2).

hh. Preparatory functions (G-codes) and miscellaneous functions (M-codes) in accordance with EIA Standard 274.

ii. Software for axis travel limits, when specified (see 6.2).

jj. Orbital interpolation, when specified (see 6.2).

kk. Programmable macros, when specified (see 6.2).

ll. Three- or four-axis simultaneous control, when specified (see 6.2).

mm. Helical interpolation, when specified (see 6.2).

oo. Mirror imaging, when specified (see 6.2).

3.4.12.1 Postprocessor. When specified (see 6.2), type II machines shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be provided for converting specified programming language to the form and format required to

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operate the machine tool and control. The postprocessor shall be written for use on the type computer specified. Documentation sufficient to implement and utilize the postprocessor shall be provided for computer compatibility, verification, and validation at least 30 days prior to delivery of the electrical discharge machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of tables I and II for the machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table III.

3.7 Performance tests. The types I and II machines shall be capable of performing as specified herein. The machines shall perform all machining operations pertinent to die sinking machines of its type, all related performance and cutting tests specified in section 4 and conform to the capacities specified in tables I and II while meeting the accuracies specified in table III.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified and fully described (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

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3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

3.13 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data ordered shall be written in the English language.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

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b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 50 to 120 degrees Fahrenheit and 5 to 95 percent relative humidity.

4.8.2 Operational test (type I). The machine shall be operated at no load for not less than 4 hours. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.2.1 Operational test (type II). The machine shall be operated under numerical control for a period of 8 hours without load. The numerical control program shall cycle each axis through its entire range, addressing all speeds, feeds, direction of rotation, and numerical control functions specified in this document. Proper operation of all machine functions, controls, motors, adjusting mechanisms, and accessories shall be verified during the test period. Should a malfunction or failure occur during operation, the test shall be restarted from setpoint and retimed from zero hours.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment

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tolerance test shall be performed prior to and subsequent to the performance tests given in 4.8.4 and 4.8.4.1. Any machine adjustments required to comply with 3.6 during or after the performance tests shall be cause for rejection.

4.8.4 Performance tests. The electro-erosion machine shall be subjected to all the tests described in 4.8.4.2 through 4.8.4.7. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.8.4.1 Maximum metal removal. Using a 0.500 inch thick tool steel workpiece and a graphite electrode having a diameter not less than 1.000 inch, the machine shall demonstrate that it is capable of eroding 0.05 cubic inch of metal per hour ampere. For the purpose of this test, the volume of the eroded metal only, excluding the volume of the core created by the coolant passage, shall be used in calculating the removal rate.

4.8.4.2 Surface finish. Using a tool steel workpiece and a metallic electrode having a diameter not less than 1.000 inch and a coolant passage not larger than 1/8 inch, and with the controls adjusted for the surface finish specified (see 6.2), the machine shall erode a spot face and produce a surface finish conforming to the requirements of the procuring activity. The above requirements are without regard to metal removal rate or amount of overcut.

4.8.4.3 Overcut test. A through hole not larger than 1.090 inches square shall be eroded through an oil-hardened tool steel workpiece not less than 0.375 inches thick. A 1.1000, $\pm$ 0.0001 inch square electrode with a coolant passage shall be used to enlarge the hole. The overcut shall be calculated for the desired surface finish. The finished hole size shall be measured at the entering side with precision gauge blocks. The overcut shall be calculated and shall be not more than 0.00075 inch per side.

4.8.4.4 Structural rigidity test. Using a steel workpiece of adequate thickness to allow precise measurements after completion of the test, the machine shall erode a through-hole using an electrode having a cross sectional area not less than 20 square inches, with the coolant pressure adjusted to not less than 30 pounds per square inch (PSI). The perpendicularity of the hole sides to the supporting surface of the workpiece shall be measured. The sides of the hole shall be perpendicular to the supporting surface within 0.0005 inch in all directions.

4.8.4.5 Small hole test. Using a tool steel workpiece not less than 0.060 inch thick and a metallic electrode having a diameter not larger than 0.100 inch and having no coolant hole, the machine shall be capable of eroding three through-holes without deformation or breakage of the electrode.

4.8.4.6 Electrode erosion control test. Using an oil hardened tool steel workpiece not less than 0.375 inch thick, and a carbon electrode having a suitable diameter, and with the controls adjusted for no wear on the electrode and for a metal removal rate not less than 50 percent of the maximum rate, the machine shall erode a through hole. The diameter of the electrode shall be measured throughout its active length and shall be not less 1 percent of its original dimensions.

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4.8.4.7 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The machines covered by this specification are intended to be used for die-sinking operations and reproducing complex shapes and cavities to a high degree of accuracy in a wide range of materials, particularly difficult to machine metals.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Type of measuring system to be used (US or SI), for manufacture of the machine, state required system, (see 3.2)

f. System of units for measuring and indicating devices (see 3.2.1).

g. Specify dial increments, if different (see 3.2.1.1).

h. Digital readout system, if required (see 3.2.1.2).

i. Specify any special digital readout system features, if required (see 3.2.1.2).

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j. Additional safety and health requirements, if required (see 3.2.5).

k. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

l. Painting, if different (see 3.3.5).

m. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

n Specify the stop and dial indicator increments, if different (see 3.4.4.3).

o. Filtering system, if required (see 3.4.5.3).

p. Electrode changer, fully describe if required (see 3.4.6).

q. Power supply and control unit, specify and fully describe (see 3.4.7)

r. Electrical system, if different (see 3.4.9).

s. Initial voltage, if different (see 3.4.9).

t. Motors, if different (see 3.4.9.1).

u. Worklight, if not required (see 3.4.10).

v. Hour meter, if required (see 3.4.11).

w. Automatic tape code recognition EIA Standards 244 and 358, if required (see 3.4.12 f).

x. Conversational programming, if required (see 3.4.12 i).

y. Scaling, if required (see 3.4.12 j).

z. Tape reader and size of feed and take up spools, if required (see 3.4.12 r).

aa. Specify part program storage length (in feet), if different (see 3.4.12 z).

bb. Parametric programming, if required (see 3.4.12 aa)

cc. Type of memory, specify and fully describe (see 3.4.12 bb).

dd. Battery backup (in hours) for controls having volatile memory, if different (see 3.4.12 cc).

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ee. Adaptive control, if required (see 3.4.12 gg).

ff. Software axis travel limits, if required (see 3.4.12 ii).

gg. Orbital interpolation, if required (see 3.4.12 jj).

hh. Programmable macros, if required (see 3.4.12 kk).

ii. Simultaneous axis control, specify number of axis to be controlled, (see 3.4.12 ll).

jj. Helical interpolation. if required (see 3.4.12 mm).

kk. Mirror imaging. if required (see 3.4.12 oo).

ll. Specify postprocessor, identify computer name and model, form and format, if required (see 3.4.12.1).

mm. Size and capacity, if different (see 3.5).

nn. Specify physical size limitation, if required (see 3.5.1).

oo. Additional performance features, if required (see 3.7.1).

pp. Specify and fully describe optional equipment, if required (see 3.9).

qq. Specify applicable repair parts, if required (see 3.10).

rr. Lubrication plate, if required (see 3.11.1).

ss. Information to be provided on lubrication plate, if different (see 3.11.1).

tt. Nameplate, if different (see 3.11.2).

uu. Inspection conditions, if different (see 4.3).

vv. First article inspection, if different (see 4.4).

ww. Quality conformance inspection, if different (see 4.5).

xx. Acceptance test, if different (see 4.6).

yy. Additional performance test, if required (see 4.8.4.7).

zz. Specify surface finish desired, (see 4.8.4.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that

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the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data , except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.13	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals.	Require all data be in the English language.
3.13	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals.	Require all data be in the English language.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International D'Units (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

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6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

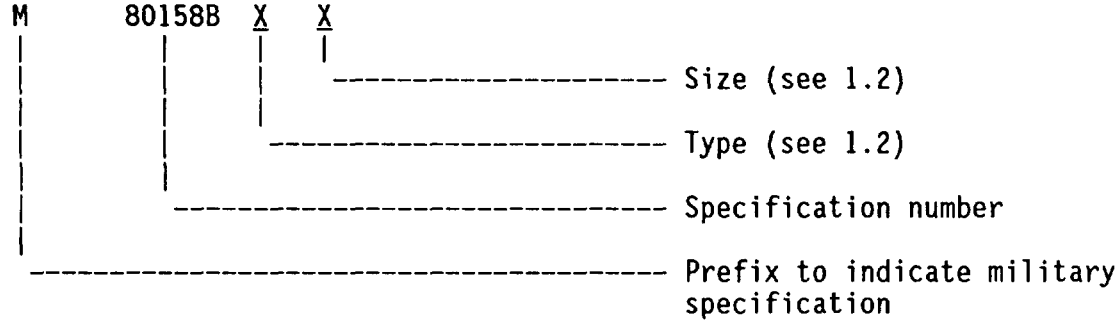
6.8 Safety and health requirements. Code of Federal Regulations 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.13 Subject term (keyword) listing.

Dielectric coolant system
Electrical discharge machine
Electrode
Platen
Postprocessor
Power supply and control unit
Ram/quill and spindle assembly
Ram/quill feed system
Table work tank

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TABLE I. Sizes and capacities.

Type I machines									
Characteristics	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8	Size 9
Worktable size (L x W), in.	15.70 x 11.80	16 x 8	24 x 12	26 x 14	27.50 x 15.80	28 x 16	36 x 16	36 x 20	38 x 20
Table workload, max. capacity, lbs.	440	750	1500	2500	1200	4000	5000	5000	2000
Electrode holder to table surface, in.	13-15	14-20	22.5	24	5-17	29	22	40	13.5-27.5
Maximum electrode wt., lbs.	22-55	250	120	200	220	300	2500	500	500
Type of electrode holder:	C20 chuck ITS(S40) chuck P100 plate 3R chuck	Platen	System 3R electromag chuck	System 3R electromag chuck	Platen	System 3R electromag chuck	Platen	System 3R electromag chuck	Platen
Axis travels:									
X axis, in.	11.8	8.0	12.5	14.0	12.0	17.0	24.0	19.0	20.0
Y axis, in.	9.8	6.0	10.0	12.0	12.0	15.0	12.0	19.0	14.0
Z axis, in.	9.8	6.0	12.0	15.0	12.0	17.0	12.0	24.0	14.0
C axis, in. (Rotary spdl)	optional attachment	N/A	optional attachment	optional attachment	N/A	optional attachment	optional attachment	optional attachment	N/A
Dielectric reservoir, capacity, gallons	40	33	55	75	80	100	115	150	225
Power supply (amperes)	32 std/ 64 opt	25 std	30 std	50 std	50 std/ 100/200 opt	75 std	30,50,75 or 100	75 std	50 std/ 100/200 opt

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TABLE II. Sizes and capacities.

Type II machines											
Characteristics	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8	Size 9	Size 10	Size 11
Worktable size, (L x W), in.	15.70 x 11.80	17.72 x 11.02	20 x 12	22.06 x 15.75	24 x 12	26 x 14	28 x 16 or 20	31.50 x 22.00	36 x 20	38 x 16	66 x 36
Table workload, max. capacity, lbs.	177	220	2000	660	400 or 1500	2200	2205 or 3500	1760	5000	4000	10000
Electrode holder to table top, max. in.	13.80	15.00 or 15.75	17	21.25 or 22.06	20 or 22.50	24	26 or 29	27 to 27.50	40.00	21.50	35.50
Max. electrode wt., lbs.	33	110	1000	440	120	200	220 or 400	1100	500	2000	4000
Type of electrode holder:	platen	platen/w opt. C20, S40 chucks &P100/150 200 plate	platen	platen/w opt. C20, S40 chucks &P100/150 200 plate	platen or electro-mag chuck	electro-mag chuck	platen or electro-mag chuck	platen/w opt. C20, S40 chucks & P100/150 200 plate	electro-mag chuck	platen	platen
Axis travels:											
X axis, in.	10.00	8.66	12.00	12.60	12.00	14.00	17 to 18	17.75	19.00	18.00	39.50
Y axis, in.	8.00	6.30	11.00	8.66	10.00	12.00	14 to 15	12.62	19.00	12.00	15.20
Z axis, in.	8.00	8.66	7.50	12.60	12.00	15.00	14 to 17	16.12	24.00	12.00	17.50
C axis, Rotary	optional	optional	optional	optional	optional	optional	optional	optional	optional	optional	optional
Auto electrd chgr:	N/A	optional	optional	optional	optional	optional	optional	optional	optional	optional	optional
Max. storage cap.	N/A	20	8,16,24 or 32	40	8 or 24	8 or 16	8 or 24	40	8,16,24 or 32	8,16,24 or 32	8,16,24, or 32
Dielectric reservior, gallons	40	45	80	77	55 or 80	75	100 or 120	148	150	135	325
Power supply (amperes)	30 or 50	64	50 std 100/200 optional	64 or 128	30,50 or 100	50	50,75 or 100	128	75	50 std/ 100/200 optional	50 std/ 100/200 optional

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TABLE III. Alignment tolerances.

Requirements	Tolerances (Not greater than)	
	Type I (Manual machine)	Type II (CNC machine)
Alignment accuracies:		
Table flatness (Concave only)	0.001" per 36 in.	0.001" per 36 in.
Rotary spindle (integral or attachment) square to table (Zero indicator and tram each 90, 180 and 270 degrees)	0.0005" per ft.	0.0005" per ft.
Ram or quill travel perpendicular to table (Use full travel)	0.0005"	0.0005"
Rise and fall of table, transversely and longitudinally, use full travel of table of each direction	0.0005" per ft. 0.0005" per ft.	0.0005" per ft. 0.0005" per ft.
Center T-slots parallel with longitudinal travel of table	0.001" per ft. 0.001" per ft.	0.001" per ft. 0.001" per ft.
Platen parallel with table surface, longitudinally and transversely	0.001" per ft.	0.001" per ft.
Electrode holder axis (taper or collet) square with table surface	0.0005"	0.0005"
Positioning accuracies:		
Linear positioning each axis		±0.00025"
Linear positioning repeatability		±0.0002"

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Custodians:
Army - AL
Navy - SH
Air Force - 99

User activities:
Air Force - 71
Navy - MC

Review activity:
Air Force - 84

Preparing activity:
DLA - IP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-E-80158B	20 Mar 90

3. DOCUMENT TITLE Electro-Erosion Machines, Electric Discharge "C" Frame, Ram/Quill Type, Stationary/Rotary Spindle, Manual and Computer Numerical Control (CNC)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center	(1) Commercial 901/775-4790	(2) AUTOVON 683-4790
c. ADDRESS (Include Zip Code) 2163 Airways Boulevard Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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